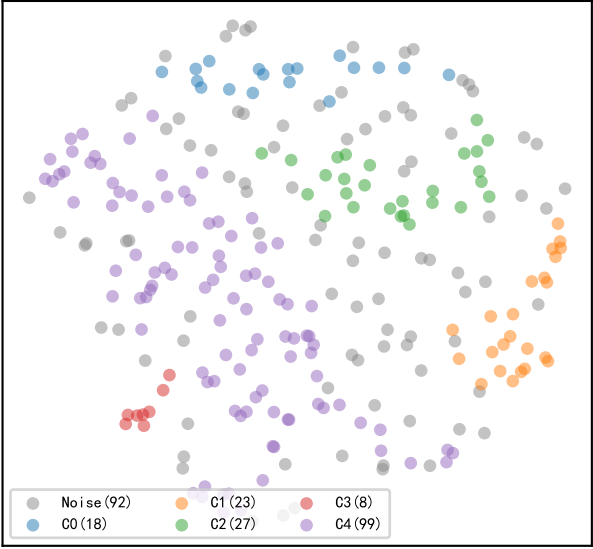
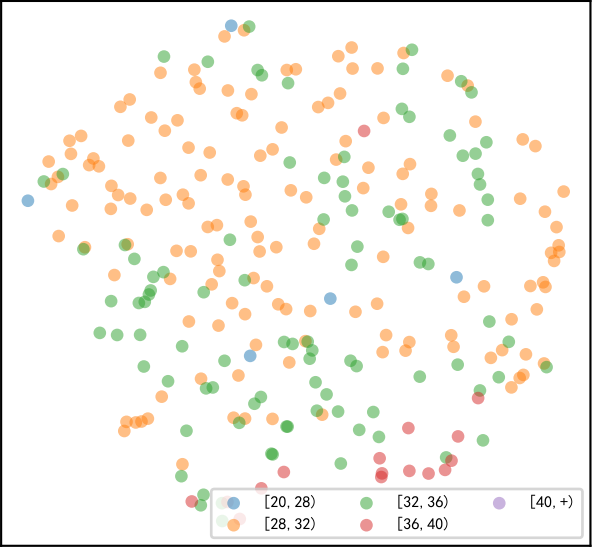


UMAP Clustering Analysis — 2025 C Problem (NIPT)

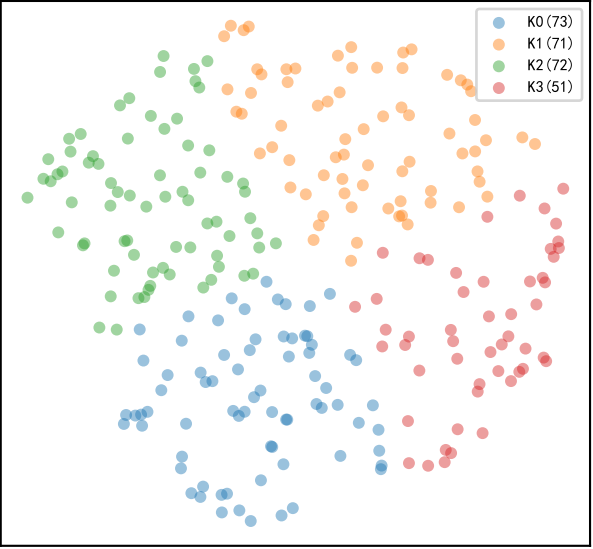
Male HDBSCAN (5+noise)



Male BMI Groups



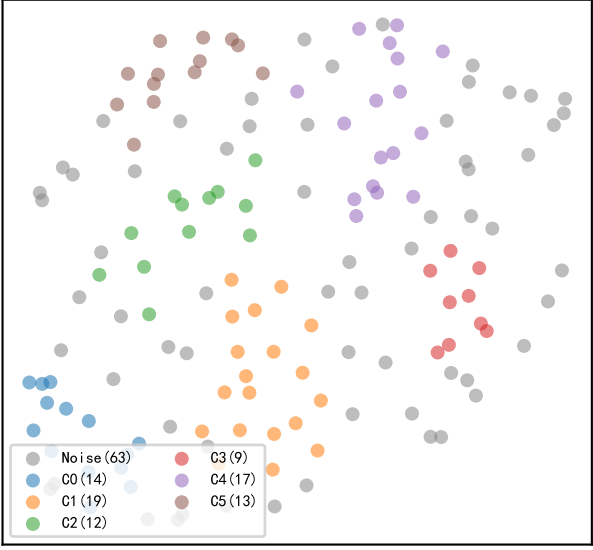
Male K-Means k=4 (Sil=0.389)



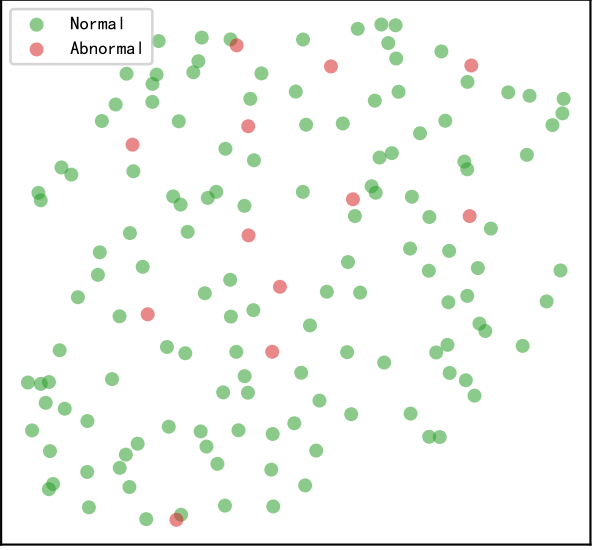
==== Male Results ====
n=267 (dedup), d=2
HDBSCAN: 5 clusters + 92 noise (34.5%)
Silhouette: HDB=0.1832
K-Means k=4 Silhouette: 0.3885

UMAP: n_neighbors=15, min_dist=0.1
Features: Y conc, BMI, GW, GC, Z-scores, tech QC

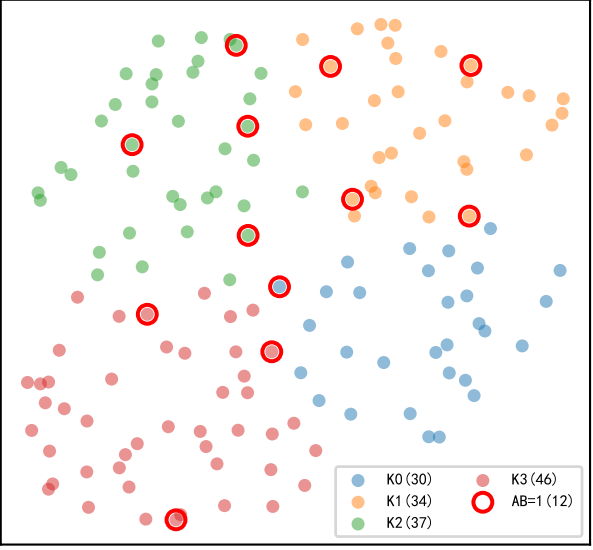
Female HDBSCAN (6+noise)



Female AB Labels (ground truth)



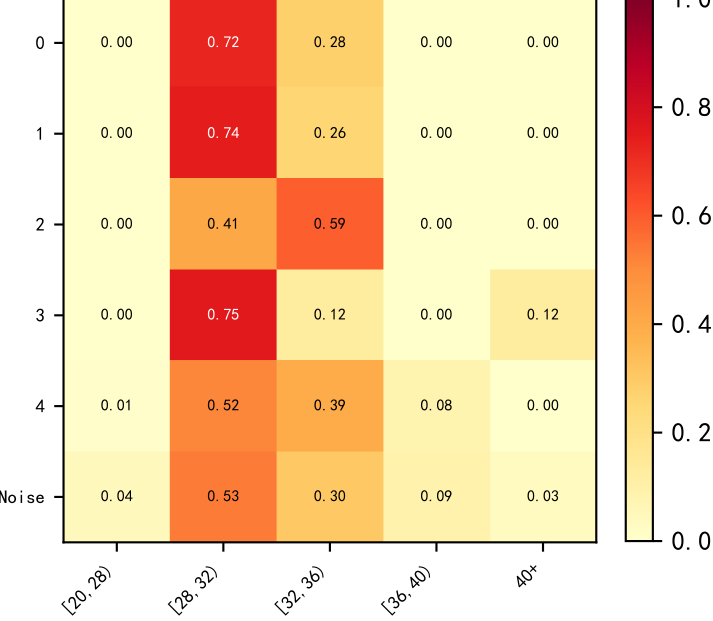
Female K-Means k=4 (Sil=0.417)



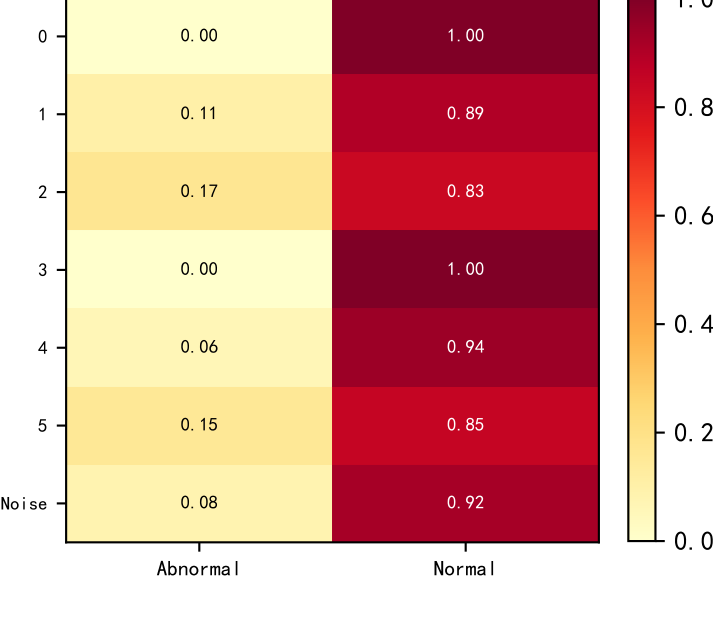
==== Female Results ====
n=147 (dedup), abnormal=12 (8.2%)
HDBSCAN: 6 clusters + 63 noise (42.9%)
Silhouette: HDB=0.5011
K-Means k=4 Silhouette: 0.4171
vs AB: HDB ARI=0.0000 NMI=0.0201
KM ARI=-0.0008 NMI=0.0090

Features: X conc, BMI, GW, GC, Z-scores, tech QC

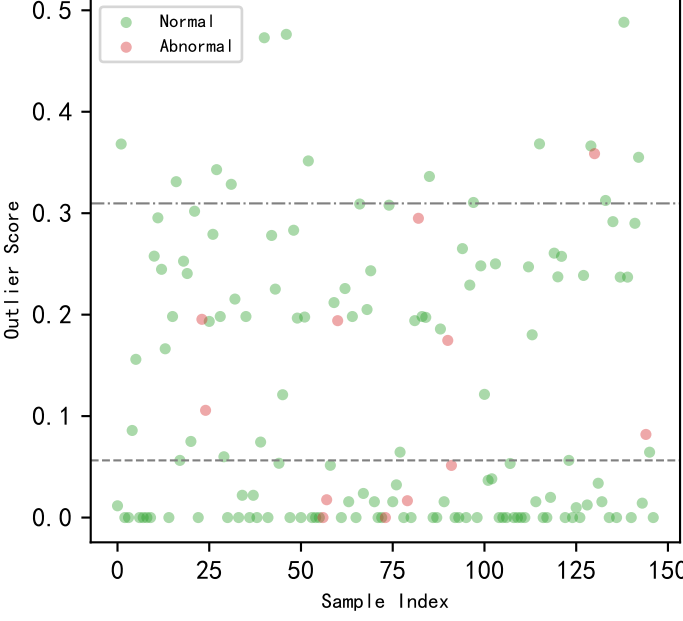
Male: Cluster -> BMI (row-%)



Female: Cluster -> AB (ARI=0.000)



Female HDBSCAN Outlier Score



Male HDBSCAN Outlier Score (by BMI)

